

Correction for "Interference as Noise: Friend or Foe?"

There is a minor mistake in the IEEE Transactions on Information Theory, Vol. 62, No. 6, pp. 3561-3596. The computation in eq.(114) is only correct in the case when $\frac{\bar{b}z_y - \epsilon}{\bar{a}z_x} \geq 1$. This is so because the bounds in eq. (117) and eq. (118) are correct. In what follows, we give the details of the correct computation of eq. (114) and show eq. (117) and eq. (118) are as in the printed version of the paper.

I. COMPUTATION OF $m(A_\epsilon^c(z_x, z_y))$ IN EQ.(114)

We begin by computing eq.(114).

Based on the values of $\epsilon, \bar{a}, \bar{b}, z_x$ and z_y the shape of the set $A_\epsilon^c(z_x, z_y)$ takes one of the following three forms:

- 1) Case 1: $\frac{\bar{b}z_y + \epsilon}{\bar{a}z_x} \leq 1$, when both edges of the strip intersect the horizontal line $b_{yy} = 1$, see Fig. 1a;
- 2) Case 2: $\frac{\bar{b}z_y - \epsilon}{\bar{a}z_x} \geq 1$, when both edges of the strip intersect the vertical line $a_{xx} = 1$, see Fig. 1b; and
- 3) Case 3: $\frac{\bar{b}z_y - \epsilon}{\bar{a}z_x} < 1$ and $\frac{\bar{b}z_y + \epsilon}{\bar{a}z_x} > 1$, when the top edge of the strip crosses the horizontal line $b_{yy} = 1$ and the bottom edge crosses the vertical line $a_{xx} = 1$, see Fig. 1c.

A. Computation for the Case 1

For the Case 1 the area of the outage strip can be computed as follows:

$$m(A_\epsilon^c(z_x, z_y)) = 1 - \underbrace{\frac{(1 - b_{yy}^{(\text{Point D})})a_{xx}^{(\text{Point A})}}{2}}_{\text{Area of the upper triangle}} - \underbrace{\frac{(1 - a_{xx}^{(\text{Point C})}) + (1 - a_{xx}^{(\text{Point B})})}{2}}_{\text{Area of the lower trapezoid}} \quad (1)$$

$$= 1 - \frac{(1 - \frac{\epsilon}{\bar{b}z_y})\frac{\bar{b}z_y - \epsilon}{\bar{a}z_x}}{2} - \frac{(1 - \frac{\bar{b}z_y + \epsilon}{\bar{a}z_x}) + (1 - \frac{\epsilon}{\bar{a}z_x})}{2} \quad (2)$$

$$= 1 - \frac{(\bar{b}z_y - \epsilon)^2}{2\bar{a}z_x\bar{b}z_y} - 1 + \frac{\bar{b}z_y}{2\bar{a}z_x} + \frac{\epsilon}{\bar{a}z_x} \quad (3)$$

$$= -\frac{\bar{b}^2z_y^2 + \epsilon^2 - 2\bar{b}z_y\epsilon}{2\bar{a}z_x\bar{b}z_y} + \frac{\bar{b}z_y}{2\bar{a}z_x} + \frac{\epsilon}{\bar{a}z_x} \quad (4)$$

$$= \frac{2\epsilon}{\bar{a}z_x} - \frac{\epsilon^2}{2\bar{a}z_x\bar{b}z_y} \quad (5)$$

$$\leq \frac{2\epsilon}{\bar{a}z_x}. \quad (6)$$

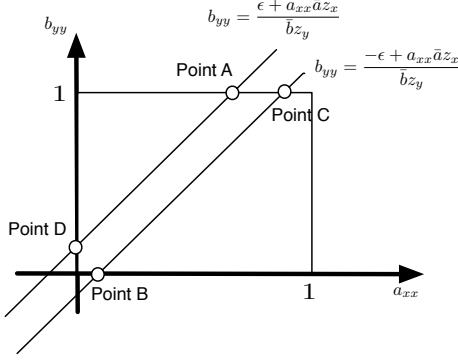
B. Computation for the Case 2

Due to symmetry for Case 2 the area is given by:

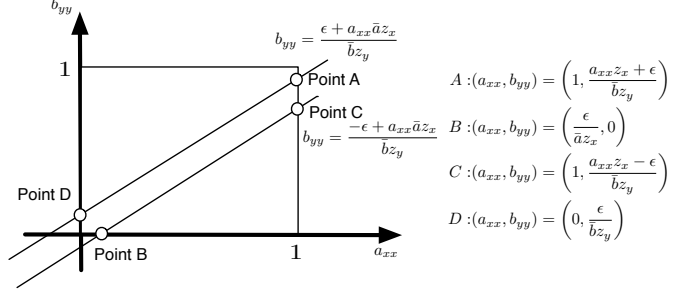
$$m(A_\epsilon^c(z_x, z_y)) = \frac{2\epsilon}{\bar{b}z_y} - \frac{\epsilon^2}{2\bar{a}z_x\bar{b}z_y} \leq \frac{2\epsilon}{\bar{b}z_y}. \quad (7)$$

Observe that the condition for Case 1 implies that $\bar{b}z_y < \bar{a}z_x$ and the condition for the Case 2 $\bar{b}z_y > \bar{a}z_x$. Therefore for these two cases the bound can be combined into

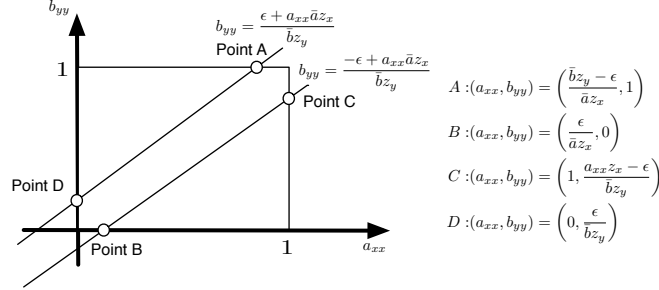
$$m(A_\epsilon^c(z_x, z_y)) \leq 2\epsilon \min\left(\frac{1}{\bar{b}z_y}, \frac{1}{\bar{a}z_x}\right). \quad (8)$$



(a) Shape of the outage strip for Case 1.



(b) Shape of the outage strip for Case 2.



(c) Shape of the outage strip for Case 3.

Fig. 1: Possible Shapes of $A_\epsilon^c(z_x, z_y)$.

C. Computation for the Case 3

For the Case 3 the area of the outage strip is given by

$$m(A_\epsilon^c(z_x, z_y)) = 1 - \underbrace{\frac{(1 - b_{yy}^{\text{Point D}})a_{xx}^{\text{Point A}}}{2}}_{\text{Area of the upper triangle}} - \underbrace{\frac{(1 - a_{xx}^{\text{Point B}})b_{yy}^{\text{Point C}}}{2}}_{\text{Area of the lower triangle}} \quad (9)$$

$$= 1 - \frac{1}{2} \left(1 - \frac{\epsilon}{\bar{b}z_y}\right) \frac{\bar{b}z_y - \epsilon}{\bar{a}z_x} - \frac{1}{2} \left(1 - \frac{\epsilon}{\bar{a}z_x}\right) \frac{\bar{a}z_x - \epsilon}{\bar{b}z_y} \quad (10)$$

$$= 1 - \frac{1}{2} \frac{(\bar{b}z_y - \epsilon)^2 + (\bar{a}z_x - \epsilon)^2}{\bar{a}z_x \bar{b}z_y} \quad (11)$$

$$= 1 - \frac{1}{2} \frac{\bar{b}^2 z_y^2 - 2\epsilon \bar{b}z_y + \epsilon^2 + \bar{a}^2 z_x^2 - 2\epsilon \bar{a}z_x + \epsilon^2}{\bar{a}z_x \bar{b}z_y} \quad (12)$$

$$= 1 - \frac{\bar{b}^2 z_y^2 + \bar{a}^2 z_x^2 + 2\epsilon^2}{2\bar{a}z_x \bar{b}z_y} + \frac{\epsilon(\bar{b}z_y + \bar{a}z_x)}{\bar{a}z_x \bar{b}z_y} \quad (13)$$

$$\stackrel{a)}{=} \frac{\epsilon(\bar{b}z_y + \bar{a}z_x)}{\bar{a}z_x \bar{b}z_y} - \frac{(\bar{b}z_y - \bar{a}z_x)^2}{2\bar{a}z_x \bar{b}z_y} - \frac{2\epsilon^2}{2\bar{a}z_x \bar{b}z_y} \quad (14)$$

$$\stackrel{b)}{\leq} \frac{\epsilon(\bar{b}z_y + \bar{a}z_x)}{\bar{a}z_x \bar{b}z_y} - \frac{2\epsilon^2}{2\bar{a}z_x \bar{b}z_y} \quad (15)$$

$$\stackrel{c)}{\leq} \min \left(\frac{\epsilon(2\bar{b}z_y + \epsilon)}{\bar{a}z_x \bar{b}z_y}, \frac{\epsilon(2\bar{a}z_x + \epsilon)}{\bar{a}z_x \bar{b}z_y} \right) - \frac{\epsilon^2}{\bar{a}z_x \bar{b}z_y} \quad (16)$$

$$= 2\epsilon \min \left(\frac{1}{\bar{a}z_x}, \frac{1}{\bar{b}z_y} \right) \quad (17)$$

where (in)-equalities follow from: a) completing the square; b) dropping the term $\frac{(\bar{b}z_y - \bar{a}z_x)^2}{2\bar{a}z_x \bar{b}z_y}$; and c) the condition of the Case 3 which implies that $\bar{b}z_y \leq \bar{a}z_x + \epsilon$ and $\bar{a}z_x \leq \bar{b}z_y + \epsilon$ and taking the tightest of the two bounds.

Therefore, for all three cases the area is bounded as follows

$$m(A_\epsilon^c(z_x, z_y)) \leq 2\epsilon \min \left(\frac{1}{\bar{a}z_x}, \frac{1}{\bar{b}z_y} \right). \quad (18)$$

II. DEMONSTRATING THAT EQ. (117) AND EQ. (118) ARE CORRECT

We now show that eq. (117) and eq. (118) are correct.

To that end observe that

$$\begin{aligned} m(A_\epsilon^c) &\leq \sum_{1 \leq z_x \leq N_x, 1 \leq z_y \leq N_y} m(A_\epsilon^c(z_x, z_y)) \\ &\leq 2\epsilon \sum_{1 \leq z_x \leq N_x, 1 \leq z_y \leq N_y} \min\left(\frac{1}{\bar{a}z_x}, \frac{1}{\bar{b}z_y}\right) \end{aligned}$$

Therefore, by using the upper bound $\min\left(\frac{1}{\bar{a}z_x}, \frac{1}{\bar{b}z_y}\right) \leq \frac{1}{\bar{b}z_y}$ we recover eq. (117)

$$m(A_\epsilon^c) \leq \frac{2\epsilon N_x(1 + \ln(N_y))}{\bar{b}},$$

and by using the upper bound $\min\left(\frac{1}{\bar{a}z_x}, \frac{1}{\bar{b}z_y}\right) \leq \frac{1}{\bar{a}z_x}$ we recover eq. (118)

$$m(A_\epsilon^c) \leq \frac{2\epsilon N_y(1 + \ln(N_x))}{\bar{a}}.$$